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## Instruction-guided visual embeddings and feedback-based learning in large vision-language models


## Abstract

In an example, a method for fine-tuning a Large Visual Language Model (LVLM) includes providing visual queries, each of the visual queries comprises at least an image and a textual query related to the image; processing, by the LVLM, the visual queries to extract visual embeddings from the visual queries, wherein the LVLM comprises a Visual Language Model (VLM), a first Large Language Model (LLM), and a linear projection layer interconnecting the VLM and the LLM; for visual queries: i) generating, by the LVLM, a response to the corresponding visual query based on the corresponding visual embedding; ii) evaluating, by a second LLM, the generated response to verify that the generated response satisfies predefined criteria; and iii) providing, by the second LLM, a feedback to the LVLM, in response to the evaluating the generated response; and fine-tuning the LVLM using aggregated feedback provided by the second LLM for the visual queries.



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## References Cited

**U.S. PATENT DOCUMENTS**

| Patent No. | Issued Date | Patentee Name | U.S. Cl. | CPC |
|---|---|---|---|---|
| 2024/0095491 | 12/2023 | Birru | N/A | H04L 51/02 |

**OTHER PUBLICATIONS**

Chen et al., “Learning to See and Align: Instruction-Guided Visual Embedding and Feedback-Based Learning”, SRI, Retrieved on: Jun. 6, 2024, 6pp. cited by applicant
Ye et al., “mPLUG-Owl: Modularization Empowers Large Language Models with Multimodality”, arXiv:2304.14178, Mar. 29, 2024, 21 pp. cited by applicant

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## Background/Summary

(1) This application claims the benefit of U.S. Patent Application 63/592,706, filed Oct. 24, 2023, which is incorporated by reference herein in its entirety.

TECHNICAL FIELD
(1) This disclosure is related to machine learning systems, and more specifically to instruction-guided visual embeddings and feedback-based learning.
BACKGROUND
(2) Large vision-language models (LVLMs) are capable of producing logical responses to user inquiries. While existing LVLMs have shown good capabilities in generating user-friendly responses based on images, their performance may be hindered by two main limitations. Existing LVLMs may often employ fixed image encoders that do not consider the instruction context when extracting visual embeddings. Static image encoding may limit flexibility and may not fully utilize the given instructions for image processing. Relying solely on instruction finetuning without human feedback may lead to the generation of hallucinated or unhelpful content. Additionally, LVLMs may be susceptible to adversarial prompting, which may elicit harmful or inappropriate

responses.

## SUMMARY

(3) In general, techniques are described for a LVLM model designed to generate human-aligned responses based on visual information and instructions. The disclosed techniques may directly connect a pretrained large language model (LLM) with a small-sized pretrained vision-language model (VLM) using a linear projection layer. The small-sized VLM may be extensively fine-tuned to extract visual embeddings based on given instructions. In some examples, the LVLM model may be initially fine-tuned on a dataset of instructions and corresponding visual information. After instruction finetuning, the LVLM model may undergo a reinforcement learning stage where it may receive feedback from an LLM.

(4) In accordance with the disclosed techniques, the feedback may consider various aspects, including, but not limited to helpfulness, honesty, and harmlessness. The LVLM model may be trained using a mix of conditional reinforcement learning and rejection sampling.

(5) The disclosed techniques may incorporate natural language feedback (NLF) from one or more LLMs. The NLF may pinpoint the strengths and weaknesses of the response of the model. In some examples, the NLF may provide concrete advice on guiding the output of the VLM model towards the ground truth.

(6) The disclosed techniques may convert NLF into a series of prompts or questions that may be used to guide system's responses in subsequent turns. This conversion may allow the disclosed LVLM to learn from the feedback and adapt its responses accordingly. As described herein, a machine learning system may modify the conditional Reinforcement Learning (RL) algorithm to incorporate NLF as a conditioning factor. The machine learning system may train the LVLM to generate responses conditioned on both the original prompt and the NLF. This may help the LVLM understand how to apply the feedback to improve its output.

(7) The techniques may provide one or more technical advantages that realize at least one practical application. For example, the disclosed techniques may improve the safety and helpfulness of vision-language models. Such improvement may be achieved by incorporating natural language feedback from LLMs into the model training process. In addition, the disclosed techniques may improve the performance of LVLM due to ability of the LVLM to effectively combine visual and textual information, and understanding of context and instructions. In some cases, conditional RL and rejection sampling may ensure that the LVLM generates high-quality responses even in challenging conditions.

(8) In an example, a method for fine-tuning a Large Visual Language Model (LVLM) includes providing a plurality of visual queries, wherein each of the plurality of visual queries comprises at least an image and a textual query related to the image; processing, by the LVLM, the plurality of visual queries to extract one or more visual embeddings from each the plurality of visual queries, wherein the LVLM comprises a Visual Language Model (VLM), a first Large Language Model (LLM), and a linear projection layer interconnecting the VLM and the LLM; for each of the plurality of visual queries: i) generating, by the LVLM, a response to the corresponding visual query based on the corresponding one or more visual embeddings; ii) evaluating, by a second LLM, the generated response to verify that the generated response satisfies one or more predefined criteria; and iii) providing, by the second LLM, a feedback to the LVLM, in response to the evaluating the generated response; and fine-tuning the LVLM using aggregated feedback provided by the second LLM for the plurality of visual queries.

(9) In an example, a computing system for generating responses by a Machine Learning (ML) system, for fine-tuning a Large Visual Language Model (LVLM), the computing system includes: processing circuitry in communication with storage media, the processing circuitry configured to execute a machine learning system comprising the LVLM, the processing circuitry configured to: provide a plurality of visual queries, wherein each of the plurality of visual queries comprises at least an image and a textual query related to the image; process, by the LVLM, the plurality of

visual queries to extract one or more visual embeddings from each the plurality of visual queries, wherein the LVLM comprises a Visual Language Model (VLM), a first Large Language Model (LLM), and a linear projection layer interconnecting the VLM and the LLM; for each of the plurality of visual queries: i) generate, by the LVLM, a response to the corresponding visual query based on the corresponding one or more visual embeddings; ii) evaluate, by a second LLM, the generated response to verify that the generated response satisfies one or more predefined criteria; and iii) provide, by the second LLM, a feedback to the LVLM, in response to the evaluating the generated response; and fine-tune the LVLM using aggregated feedback provided by the second LLM for the plurality of visual queries.

(10) In an example, non-transitory computer-readable storage media having instructions encoded thereon, the instructions configured to cause processing circuitry to: provide a plurality of visual queries, wherein each of the plurality of visual queries comprises at least an image and a textual query related to the image; process, by a Large Visual Language Model (LVLM), the plurality of visual queries to extract one or more visual embeddings from each the plurality of visual queries, wherein the LVLM comprises a Visual Language Model (VLM), a first Large Language Model (LLM), and a linear projection layer interconnecting the VLM and the LLM; for each of the plurality of visual queries: i) generate, by the LVLM, a response to the corresponding visual query based on the corresponding one or more visual embeddings; ii) evaluate, by a second LLM, the generated response to verify that the generated response satisfies one or more predefined criteria; and iii) provide, by the second LLM, a feedback to the LVLM, in response to the evaluating the generated response; and fine-tune the LVLM using aggregated feedback provided by the second LLM for the plurality of visual queries.

(11) The details of one or more examples of the techniques of this disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques will be apparent from the description and drawings, and from the claims.

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## Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a block diagram illustrating an example computing environment for a LVLM employing instruction-guided visual embeddings and feedback-based learning, in accordance with one or more techniques of the disclosure.

(2) FIG. **2** is a detailed block diagram illustrating an example computing system, in accordance with the techniques of the disclosure.

(3) FIG. **3** illustrates an example architecture of a LVLM model, in accordance with the techniques of this disclosure.

(4) FIG. **4** illustrates an example of a helpful output response that may be generated by a LVLM model, in accordance with the techniques of the disclosure.

(5) FIG. **5** illustrates an example of a harmful output response that may be generated by a LVLM model.

(6) FIG. **6** is a detailed block diagram illustrating an example of Natural Language Feedback (NLF) annotation performed by a LLM, in accordance with the techniques of the disclosure.

(7) FIG. **7** is a flowchart illustrating an example mode of operation for a machine learning system, according to techniques described in this disclosure.

(8) Like reference characters refer to like elements throughout the figures and description.

DETAILED DESCRIPTION

(9) While LVLMs have shown useful capabilities in generating user-friendly responses based on images, performance of LVLMs may be hindered by two main limitations. Existing LVLMs often use fixed image encoders that do not consider the instruction context when extracting visual

embeddings. Static image encoding may limit flexibility and may not fully utilize the given instructions for image processing. Relying solely on instruction finetuning without human feedback may lead to the generation of hallucinated or unhelpful content. Additionally, conventional LVLMs may be susceptible to adversarial prompting, which may elicit harmful or inappropriate responses.

(10) The disclosed techniques provide a large vision-language model (LVLM) designed to overcome the limitations of existing LVLMs. The disclosed LVLM may directly connect a pretrained large language model (LLM) with a small-sized pretrained Vision-Language Model (VLM) using a linear projection layer.

(11) As noted above, the small-sized VLM may be extensively fine-tuned to extract visual embeddings based on given instructions. As used herein, the term "visual embeddings" refers to embeddings that represent the visual content of the image in a numerical format. For example, if the visual query is an image of a cat, the VLM may generate a visual embedding that captures the features of the cat (e.g., fur color, shape, tail length). Instruction-guided visual embedding may allow the disclosed LVLM to consider the instruction context when processing images, addressing the limitation of static image encoding. The LVLM model may be initially fine-tuned on a dataset of instructions and corresponding visual information.

(12) After instruction finetuning, the LVLM model may undergo a reinforcement learning stage where the LVLM model may receive feedback from an LLM. The LLM may be trained to judge the responses of the LVLM model and to provide both scalar rewards and textual feedback. The feedback may consider various aspects, including, but not limited to, helpfulness, honesty, and harmlessness. The LVLM may be trained using a mix of conditional reinforcement learning and rejection sampling.

(13) The performance of LVLM model may be evaluated on open-domain visual question answering task and resistance to adversarial prompting task. The open-domain visual question answering task may involve answering questions about images without relying on specific knowledge or training data. The disclosed LVLM model has shown improved performance compared to previous LVLMs on this task, as evidenced by both automatic and human evaluation. Resistance to adversarial prompting task may involve evaluating the ability of the LVLM model to withstand malicious attempts to elicit harmful or inappropriate responses. The disclosed LVLM has demonstrated greater resilience to adversarial prompting than previous LVLMs, suggesting that the disclosed training paradigm and feedback mechanism may effectively mitigate this vulnerability.

(14) The small-sized VLM may be extensively fine-tuned to extract visual embeddings based on given instructions. Instruction-guided visual embedding may allow the LVLM to consider the instruction context when processing images, addressing the limitation of static image encoding. The LVLM model may be initially fine-tuned on a dataset of instructions and corresponding visual information. After instruction finetuning, the LVLM may be trained using a mix of conditional reinforcement learning and rejection sampling.

(15) The disclosed techniques involve extracting visual embeddings that are context-aware and consider the given instructions. Instruction-guided visual embedding extraction may be achieved by fine-tuning a small-sized VLM with a linear projection layer, allowing the LVLM model to adapt its visual embedding extraction based on the specific instructions

(16) The disclosed techniques may allow the LVLM to interface with instructions and extract visual embeddings that align with these instructions.

(17) Advantageously, instruction-guided visual embedding extraction may provide a significant improvement over the static approaches used in previous LVLMs. The small-sized VLM and the linear projector may be extensively fine-tuned during the training process, enabling the model to adapt and optimize the visual embedding extraction process.

(18) The disclosed techniques may represent and process information from various modalities (like images, audio, etc.) through a common language-based interface. Expansibility may allow the LVLM to leverage pretrained LLMs and a wide range of pretrained multimodal models (X-

Language Models), making it adaptable to different tasks and modalities. By representing all information in a language-based format, the disclosed techniques may more easily integrate and process information from different sources.
(19) FIG. **1** is a block diagram illustrating example computing environment **10** for a LVLM employing instruction-guided visual embeddings and feedback-based learning, in accordance with one or more techniques of the disclosure. Computing environment **10** includes computing system **100**, computing device **150**, training data **122**, and network **111**. Computing device **150** may include a mobile computing device, such as a mobile phone (including a smartphone), a laptop computer, a tablet computer, a wearable computing device, or any other computing device. In the example of FIG. **1**, computing device **150** stores multimodal data **152** and graphical user interface (GUI) **154**. Multimodal data **152** may include a plurality of images from the real world, such as, but not limited to, photos, paintings, or illustrations. A plurality of images in the multimodal data **152** may depict a wide range of scenes, objects, and people. Multimodal data **152** may also include one or more synthetic images. For example, synthetic images may be images generated by computer programs, often using techniques like 3D rendering or generative models. Synthetic images may be used to create controlled datasets with specific characteristics.
(20) GUI **154** may include a user interface associated with functionality of computing device **150**. For example, GUI **154** of FIG. **1** may be a user interface for a software application associated with a LVLM, such as LVLM **102**. Although illustrated in FIG. **1** as internal to computing device **150**, GUI **154** may generate output for display on an external display device. In some examples, GUI **154** may provide an option for a user of computing device **150** to input multimodal data **152**, such as visual data (images) and textual data (prompt/query). GUI **154** may provide an option for a user of computing device **150** to input multimodal data **152** to LVLM **102**, which may output textual descriptions, answers to visual queries, or generated images based on text included in multimodal data **152**. Although described as a graphical user interface, GUI **154** may represent any type of interface by which a user of computing device **150** can perform operations attributed herein to GUI **154**, such as a command line interface, a website or form, or some combination thereof.
(21) Although illustrated as external to computing system **100**, computing device **150** may be a component of computing system **100**. Computing device **150** and computing system **100** may communicate via communication channel **111**, which may include a network, such as the Internet, another public or private communications network, for instance, broadband, cellular, Wi-Fi, ZigBee, Bluetooth® (or other personal area network—PAN), Near-Field Communication (NFC), ultrawideband, satellite, enterprise, service provider and/or other types of communication networks, or other types of communication channels for transmitting data between computing systems, servers, and computing devices. Alternatively, or in addition, although not shown, computing system **100** may receive multimodal data **152** from a storage device that interfaces with computing system **100** and that stores multimodal data **152**. Such storage devices may include a USB drive, a disk drive (e.g., solid state drive or hard drive), an optical disc, or other storage device or media.
(22) Computing system **100** may represent one or more computing devices configured to execute LVLM **102**. LVLM **102** may comprise VLM **112**, one or more linear projectors **114** (also referred to herein as "linear projection layers **114**"), and LLM **116**. VLM **112** may include computer-readable instructions for understanding and processing both visual and textual information simultaneously. For example, VLM **112** may be trained on massive datasets of images and text, allowing VLM **112** to learn relationships between the two modalities. Small-sized VLM **112** may offer a more practical solution, as compared to large-scale VLMs, especially for devices with limited resources.
(23) Traditional LVLMs typically employ a fixed image encoder to extract visual features from images. The fixed image encoder is often pretrained on large-scale datasets and remains unchanged during downstream tasks. The extracted visual features are then combined with textual information to perform various tasks. LVLM **102**, on the other hand, introduces a more flexible architecture.

LVLM **102** may incorporate linear projection layer **114** that may be fine-tuned along with the VLM **112**. The linear projection layer **114** may act as a bridge between the visual features extracted by the fixed encoder and the VLM **112**. Training data **122** may include a storage device configured to store at least feedback data, as described below.

(24) LLM **116** may include computer-readable instructions for understanding and generating human language. LLM **116** may be trained on massive amounts of text data, which may allow LLM **116** to learn patterns, grammar, and semantics. LLM **116** may be trained on billions or even trillions of words, making LLM **116** very powerful. LLM **116** may be trained to generate a response to a visual query inputted by a user.

(25) Computing system **100** may receive a request from computing device **150** to perform a query on multimodal data **152**. Computing device **150** may output GUI **154** as a user interface with options for a user operating computing device **150** to input multimodal data **152**. For example, computing device **150** may output GUI **154** that allows a user to input an image and input a question related to the image. In some instances, computing device **150** may retrieve, via network **111**, at least a portion of multimodal data **152** (e.g., image data) from external sources such as the Internet. Computing device **150** may send, via network **111**, multimodal data **152**, which may include image data and one or more queries related to the image data, to computing system **100**.

(26) In accordance with the techniques described herein, computing system **100** may perform analysis of multimodal data **152**. Computing system **100** may process multimodal data **152** to generate a response to the query inputted by a user. Computing system **100** may generate corresponding responses to user queries conditioned on NLF and, in some examples, account for human preferences, as described below in conjunction with FIG. **1**.

(27) In an example, a plurality of visual queries may be provided to LVLM **102**. Each of the plurality of visual queries may include at least an image and a textual query related to the image. The following steps may be repeated for each of the plurality of the provided visual queries. The VLM **112** component of LVLM **102** may process the image to extract visual features. The output of VLM **112** may be passed through the linear projection layer **114** to align it with the input space of LLM **116**. The combined visual and textual embeddings may be fed into the LLM **116**. The LLM **116** may generate a textual response based on the processed visual query. A second LLM **216** (shown in FIG. **2**) may evaluate the generated response against predefined criteria, such as helpfulness, honesty, and harmlessness. The second LLM **216** may provide feedback to the LVLM **102**, indicating whether the response meets the criteria or needs improvement. The feedback may be in the form of scores, critique, and/or specific refinement suggestions. The aggregated feedback provided by the second LLM **216** for the plurality of visual queries may be used to fine-tune the LVLM **102**. The feedback loop may help ensure that the generated responses are relevant, coherent, and accurate.

(28) FIG. **2** is a block diagram illustrating an example computing system **200**. In an aspect, computing system **200** may represent computing system **100** shown in FIG. **1**. Recent advancements in AI have witnessed a significant shift from traditional LLMs to Large Vision Language Models (LVLMs). LVLMs are capable of understanding and generating both text and visual information, making LVLMs valuable for a wide range of applications. As shown, computing system **200** includes processing circuitry **243** and memory **202** for executing a machine learning system **204** having a LVLM model **102** comprising a VLM **112**, linear projection layer **114** and LLM **116**. The LVLM model **102** may include any one or more of various types of LVLM models, such as, but not limited to, generative models, modular models, knowledge-based models, multi-modal models and the like.

(29) Computing system **200** may be implemented as any suitable computing system, such as one or more server computers, workstations, laptops, mainframes, appliances, cloud computing systems, High-Performance Computing (HPC) systems (i.e., supercomputing) and/or other computing systems that may be capable of performing operations and/or functions described in accordance

with one or more aspects of the present disclosure. In some examples, computing system **200** may represent a cloud computing system, a server farm, and/or server cluster (or portion thereof) that provides services to client devices and other devices or systems. In other examples, computing system **200** may represent or be implemented through one or more virtualized compute instances (e.g., virtual machines, containers, etc.) of a data center, cloud computing system, server farm, and/or server cluster. In some examples, at least a portion of system **200** is distributed across a cloud computing system, a data center, or across a network, such as the Internet, another public or private communications network, for instance, broadband, cellular, Wi-Fi, ZigBee, Bluetooth® (or other personal area network—PAN), Near-Field Communication (NFC), ultrawideband, satellite, enterprise, service provider and/or other types of communication networks, for transmitting data between computing systems, servers, and computing devices.

(30) The techniques described in this disclosure may be implemented, at least in part, in hardware, software, firmware or any combination thereof. For example, various aspects of the described techniques may be implemented within processing circuitry **243** of computing system **200**, which may include one or more of a microprocessor, a controller, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or equivalent discrete or integrated logic circuitry, or other types of processing circuitry. Processing circuitry **243** of computing system **200** may implement functionality and/or execute instructions associated with computing system **200**. Computing system **200** may use processing circuitry **243** to perform operations in accordance with one or more aspects of the present disclosure using software, hardware, firmware, or a mixture of hardware, software, and firmware residing in and/or executing at computing system **200**. The term “processor” or “processing circuitry” may generally refer to any of the foregoing logic circuitry, alone or in combination with other logic circuitry, or any other equivalent circuitry. A control unit comprising hardware may also perform one or more of the techniques of this disclosure.

(31) Memory **202** may comprise one or more storage devices. One or more components of computing system **200** (e.g., processing circuitry **243**, memory **202**) may be interconnected to enable inter-component communications (physically, communicatively, and/or operatively). In some examples, such connectivity may be provided by a system bus, a network connection, an inter-process communication data structure, local area network, wide area network, or any other method for communicating data. The one or more storage devices of memory **202** may be distributed among multiple devices.

(32) Memory **202** may store information for processing during operation of computing system **200**. In some examples, memory **202** comprises temporary memories, meaning that a primary purpose of the one or more storage devices of memory **202** is not long-term storage. Memory **202** may be configured for short-term storage of information as volatile memory and therefore not retain stored contents if deactivated. Examples of volatile memories include random access memories (RAM), dynamic random-access memories (DRAM), static random access memories (SRAM), and other forms of volatile memories known in the art. Memory **202**, in some examples, may also include one or more computer-readable storage media. Memory **202** may be configured to store larger amounts of information than volatile memory. Memory **202** may further be configured for long-term storage of information as non-volatile memory space and retain information after activate/off cycles. Examples of non-volatile memories include magnetic hard disks, optical discs, Flash memories, or forms of electrically programmable memories (EPROM) or electrically erasable and programmable (EEPROM) memories. Memory **202** may store program instructions and/or data associated with one or more of the modules described in accordance with one or more aspects of this disclosure.

(33) Processing circuitry **243** and memory **202** may provide an operating environment or platform for one or more modules or units (e.g., LVLM model **102**, LLM model **116**, linear projection layer **114**, VLM model **112**), which may be implemented as software, but may in some examples include

any combination of hardware, firmware, and software. Processing circuitry **243** may execute instructions and the one or more storage devices, e.g., memory **202**, may store instructions and/or data of one or more modules. The combination of processing circuitry **243** and memory **202** may retrieve, store, and/or execute the instructions and/or data of one or more applications, modules, or software. The processing circuitry **243** and/or memory **202** may also be operably coupled to one or more other software and/or hardware components, including, but not limited to, one or more of the components illustrated in FIG. **2**.

(34) Processing circuitry **243** may execute machine learning system **204** using virtualization modules, such as a virtual machine or container executing on underlying hardware. One or more of such modules may execute as one or more services of an operating system or computing platform. Aspects of machine learning system **204** may execute as one or more executable programs at an application layer of a computing platform.

(35) One or more input devices **244** of computing system **200** may generate, receive, or process input. Such input may include input from a keyboard, pointing device, voice responsive system, video camera, biometric detection/response system, button, sensor, mobile device, control pad, microphone, presence-sensitive screen, network, or any other type of device for detecting input from a human or machine.

(36) One or more output devices **246** may generate, transmit, or process output. Examples of output are tactile, audio, visual, and/or video output. Output devices **246** may include a display, sound card, video graphics adapter card, speaker, presence-sensitive screen, one or more USB interfaces, video and/or audio output interfaces, or any other type of device capable of generating tactile, audio, video, or other output. Output devices **246** may include a display device, which may function as an output device using technologies including liquid crystal displays (LCD), quantum dot display, dot matrix displays, light emitting diode (LED) displays, organic light-emitting diode (OLED) displays, cathode ray tube (CRT) displays, e-ink, or monochrome, color, or any other type of display capable of generating tactile, audio, and/or visual output. In some examples, computing system **200** may include a presence-sensitive display that may serve as a user interface device that operates both as one or more input devices **244** and one or more output devices **246**.

(37) One or more communication units **245** of computing system **200** may communicate with devices external to computing system **200** (or among separate computing devices of computing system **200**) by transmitting and/or receiving data, and may operate, in some respects, as both an input device and an output device. In some examples, communication units **245** may communicate with other devices over a network. In other examples, communication units **245** may send and/or receive radio signals on a radio network such as a cellular radio network. Examples of communication units **245** may include a network interface card (e.g., such as an Ethernet card), an optical transceiver, a radio frequency transceiver, a GPS receiver, or any other type of device that can send and/or receive information. Other examples of communication units **245** may include Bluetooth®, GPS, 3G, 4G, and Wi-Fi® radios found in mobile devices as well as Universal Serial Bus (USB) controllers and the like.

(38) In the example of FIG. **2**, machine learning system **204** may receive input data from an input data set **210** and may generate output data **212**. Input data **210** and output data **212** may contain various types of information, which will generally be tailored to the application/use case for the LVLM models. When used in the example system of FIG. **1**, input data **210** may include an image. Other types of input data **210** may include various types of text including a prompt related to an image, a visual instruction, multi-modal data, and so forth. Output data **212** may include information such as, but not limited to, response to a prompt, question, visual instruction, and the like.

(39) Machine learning system **204** may process training data **213** to train the LVLM **102**, in accordance with techniques described herein. For example, a pretraining stage may involve training LVLM **102** on a massive amount of text and image data. The goal of the pretraining stage may be

for LVLM **102** to learn general representations of language and vision. As other examples, training data **213** may come in the form of billions or even trillions of words, making the process of training LLM **116** of LVLM **102** very powerful. During training, machine learning system **204** may employ a two-stage training process.

(40) As noted above, LLM **116** may be a large language model that has been trained on massive amounts of text data. LLM **116** may provide LVLM model **102** with a deep understanding of language and context. Small-sized pretrained VLM **112** may be a vision-language model that has been trained on a smaller dataset of images and text. VLM **112** may be more efficient computationally than larger models and may be well-suited for tasks that require real-time processing. Linear projection layer **114** may be a layer that connects the LLM **116** and the VLM **112**. Linear projection layer **114** may be responsible for projecting the visual features (embeddings) extracted by the VLM **112** into a space that is compatible with the language embeddings of the LLM **116**. The VLM **112** may process the input image and may extract one or more visual features of the input image.

(41) Advantageously, the extracted visual features may be passed through the linear projection layer **114**. According to the disclosed techniques, linear projection layer **114** may transform the extracted visual features into a space that is compatible with the language embeddings of the LLM **116**. In one non-limiting example, the LLM **116** may process the input instruction and may generate a language embedding.

(42) The projected visual features may be combined with the language embedding by linear projection layer **114** to create a joint representation. A byproduct of the disclosed techniques is that the joint representation may be used to perform the desired task, such as image captioning or visual question answering, as described below.

(43) In an example, unlike conventional LVLMs that use fixed image encoders, architecture of LVLM **102** may allow for extensive fine-tuning of the small-sized VLM **112** with the linear projection layer **114**. In other words, LVLM **102** may adapt its visual embeddings to the specific instructions provided, leading to more accurate and relevant results.

(44) In one non-limiting example, by fine-tuning the linear projection layer **114**, LVLM **102** may learn to extract visual features that are more aligned with the given instruction. Instruction-guided visual embedding extraction may help to mitigate the limitations of static visual embeddings that may not always be relevant to the task at hand.

(45) As noted above, the machine learning system **204** may employ a Reinforcement Learning from LLM (AI) Feedback (RLAIF) training paradigm that may further refine the ability of LVLM **102** to generate high-quality responses that are aligned with human preferences. In an aspect, machine learning system **204** may use human-annotated dense captions as ground truth examples to evaluate the generated responses of LVLM **102**. In the example illustrated in FIG. **2**, the machine learning system **204** may employ one or more LLMs **216** that may evaluate the generated responses of the LVLM **102** based on these human-annotated captions. During evaluation of the generated responses, LLM **216** may provide both scalar rewards (e.g., 0-1 scores) and textual feedback (e.g., comments on helpfulness, honesty, and harmlessness). The VLM **112** and/or the LLM **116** may be updated based on the feedback of LLM **216**, using reinforcement learning techniques to improve behavior of LVLM **102**. The evaluation of LLM **116** of fine-grained attributes may help to identify and address issues like hallucinations and lack of helpfulness. Conditional RL technique may allow LVLM **102** to learn to generate responses that are conditioned on specific inputs (e.g., images and instructions).

(46) One of the tasks performed by LVLM **102** may involve answering questions about images without relying on specific training data **213**. Advantageously, LVLM **102** may outperform conventional LVLMs on this task. Such improvement in the performance may be due to ability of LVLM **102** to effectively combine visual and textual information, and understanding of context and instructions. Adversarial prompting involves designing inputs that are intended to mislead or

confuse the model. Essentially, LVLM **102** may show a higher degree of resistance to adversarial prompting compared to exiting LVLMs. Such resilience of LVLM **102** may be due to robust training paradigm of LVLM **102** and ability of LVLM **102** to learn from human feedback.

(47) Metrics like, but not limited to, accuracy, F1-score, and BLEU (BiLingual Evaluation Understudy) score may be used to quantitatively assess the performance of LVLM **102**.

(48) The core idea of the disclosed techniques is to address the shortcomings of previous models through two key design elements: architectural innovations and training paradigm improvements. Unlike traditional LVLMs that use fixed image encoders, the disclosed techniques incorporate linear projection layer **114** that may be fine-tuned along with the small-sized VLM **112**. Dynamic image encoding may allow LVLM **102** to adapt visual embeddings to specific instructions, improving alignment and context understanding. Architecture of LVLM **102** provides a direct interface of the pretrained LLM **116** with the small-sized pretrained VLM **112**. Such interface may facilitate a seamless flow of information between the language and vision components, enhancing the ability of LVLM **102** to understand and generate coherent responses. LVLM **102** may employ an RLAIF stage after instruction fine-tuning. Human-annotated dense captions may be used to evaluate the responses of VLM **112**, providing the VLM **112** with more accurate feedback. RLAIF may help LVLM **102** align with human preferences and reduce hallucinations. LVLM **102** may use a combination of conditional RL and rejection sampling to improve robustness of the model against adversarial prompting and other vulnerabilities. In an example, conditional RL and rejection sampling may ensure that LVLM **102** generates high-quality responses even in challenging conditions.

(49) FIG. **3** illustrates an example architecture of a LVLM model, in accordance with the techniques of this disclosure. LVLM **102** marks a departure from traditional LVLMs by introducing instruction-guided visual embedding extraction.

(50) Many conventional LVLMs like BLIP2 (Bootstrapping Language-Image Pretraining), LLaVa (Language-to-Image Visual Assistant), InstructBLIP, and miniGPT4 (Generative Pre-trained Transformer) typically follow a two-step training process. The LLM may be trained on a massive dataset of image-caption pairs. A powerful vision model (e.g., CLIP, ViT) may be used to extract visual features from the images. The LLM may be further fine-tuned on a dataset of visual instructions and corresponding responses. Task-specific adaptation step may help the LLM to learn how to follow specific visual instructions and generate relevant responses, such as response shown in FIG. **4**.

(51) BLIP2 models are known for strong performance in image captioning and visual question answering, BLIP2 may leverage the power of CLIP for visual feature extraction. A multimodal foundation model, LLaVa excels at various tasks such as image captioning, visual question answering, and text-to-image generation. Instruct BLIP is a LVLM model that is specifically designed for instruction following tasks, making it particularly effective for applications like visual search and object recognition. miniGPT4 is a smaller and more efficient version of GPT-4. The miniGPT4 model offers a good balance between performance and computational cost.

(52) LVLMs have opened up new possibilities in various fields, including, but not limited to: computer vision, natural language processing and robotics. The LVLM techniques may help with image captioning, object detection, and visual search tasks of computer vision. LVLM models may also help with machine translation, summarization, and text generation of natural language processing tasks. Finally, LVLM models may also assist with visual navigation, object manipulation, and human-robot interaction tasks of robotics.

(53) While conventional LVLMs have made significant strides, there are still two primary limitations that need to be addressed. LVLMs may sometimes generate responses that are irrelevant, inaccurate, or even harmful. This misalignment with human preferences may lead to negative user experiences and undermine the credibility of LVLM model.

(54) Conventional LVLMs may extract visual embeddings without considering the specific context

or instructions provided. Context-agnostic encoding may lead to misalignment between the extracted features and the desired task. The fixed nature of the image encoder may restrict the ability of LVLM to adapt to different instructions and scenarios.

(55) In contrast, architecture of LVLM **102** illustrated in FIG. **3** may allow LVLM **102** to dynamically adjust its visual embeddings based on the provided instructions **304**. Instruction-guiding encoding may ensure that the extracted features **306** are relevant to the task at hand.

(56) LVLM **102** employs novel techniques to LVLMs by combining a pretrained LLM **116** with a small-sized VLM **112** through linear projection layer **114**. The architecture illustrated in FIG. **3** addresses the limitations of existing LVLM models by: instruction-guided visual embedding extraction and enhanced flexibility and adaptability. The linear projection layer **114** may allow the VLM model **112** to interface with instructions **304**, guiding the extraction of visual embeddings **306** that are relevant to the given task. Instruction-guided visual embedding **302** extraction is a significant improvement over the static methods used in previous LVLMs. By training the small-sized VLM **112** and the linear projector **114** together, LVLM **102** may adapt and optimize the visual embedding extraction process. Such training paradigm may make the LVLM **102** more flexible and adaptable to different tasks and instructions.

(57) In accordance with the techniques of this disclosure, the foundational design of LVLM **102** is based on the principle of "language as the unified interface." In other words, LVLM **102** may use language as a common medium to represent and process information from different modalities. By focusing on different modalities, LVLM **102** may transcribe various modalities (e.g., images, audio) into the language modality. Modality transcription may allow LVLM **102** to leverage the powerful capabilities of LLMs for understanding and processing information from these different sources. The "language as the unified interface" techniques may make LVLM **102** highly expandable. LVLM **102** may easily incorporate pretrained LLMs **116** and X-Language Models (models trained on different modalities) to handle new tasks or modalities.

(58) By using language as a common medium, LVLM **102** may leverage the efficiency and effectiveness of LLM **116** for various tasks. The ability of LVLM **102** to incorporate different X-Language Models may make LVLM **102** highly adaptable to new challenges and domains.

(59) The disclosed techniques introduce a three-stage training paradigm to address the limitations of previous LVLMs: pretraining, Supervised Fine-Tuning (SFT) and RL.

(60) This initial pretraining stage may involve training LVLM **102** on a massive amount of text and image data. The goal of the pretraining stage may be for LVLM **102** to learn general representations of language and vision.

(61) In an example, in the SFT stage, LVLM **102** may be fine-tuned on specific tasks using labeled datasets. In one example, the SFT may help LVLM **102** learn to perform specific tasks like image captioning or visual question answering. Advantageously, as used in the disclosed techniques, RL may further calibrate the responses **308** of LVLM **102** by incorporating human feedback. Human experts may provide feedback on the generated responses **308** of LVLM **102**.

(62) This feedback provided by humans may include, but is not limited to, ratings, comments, or specific examples of incorrect or harmful content. A reward function may be defined to quantify the quality of the responses **308** of LVLM **102** based on the human feedback. This reward function may be based on metrics like, but not limited to, accuracy, relevance, and helpfulness. LVLM **102** may be trained using reinforcement learning techniques to maximize the reward function. Such learning may involve iteratively generating responses **308**, receiving feedback, and updating the parameters of LVLM **102**. By incorporating human feedback through reinforcement learning, LVLM **102** may better align responses **308** with human preferences and values. The RL stage may help to identify and mitigate harmful, hallucinated, and unhelpful content in the responses **308** of LVLM **102**.

(63) In one non-limiting example, machine learning system **204** may employ LLaVA multimodal instruction tuning suite and GPT-4. The LLaVA multimodal instruction tuning suite may provide a

rich dataset of images **310** paired with corresponding instructions **304** and questions **312**. The images **310** in LLaVA may be sourced from the COCO dataset, which may include detailed annotations. The COCO dataset allow LLM **116** to effectively understand the visual content. GPT-4, a powerful LLM **216**, may be used to provide feedback on the generated responses **308** of LVLM **102**. The feedback of GPT-4 may be designed to mimic human judgments. The detailed annotations in the COCO dataset may enable GPT-4 to effectively understand the images **310**. Previous studies have demonstrated the effectiveness of LLMs for annotation tasks. Using GPT-4 for feedback may be more cost-effective than relying solely on human annotations. GPT-4 may provide additional feedback that complements human annotations, further improving performance of LVLM **102**.
(64) In one non-limiting example, D.sub.r: may be a subset of the LLaVA suite that may be set aside for training the LVLM model **102**. The D.sub.r: may ensure that the LVLM model **102** is trained on a separate dataset from the one used to fine-tune the LVLM model **102**. In this example, three distinct LVLM models **102** may be used. Each LVLM model **102** may be initialized with a different checkpoint, varying the pretraining steps. Separate checkpoints may introduce diversity and may help to avoid overfitting.
(65) All three models may undergo supervised fine-tuning (SFT) on the LLaVA suite. For each instruction and image in D.sub.r, all three LVLM models **102** may generate responses **308**. Instead of using pairwise comparison, LLM **216** (GPT-4 in this case) may be prompted to provide feedback for each individual response **308**. It should be noted that LLM **216** feedback may allow for a more nuanced evaluation of the responses **308**.
(66) Training three distinct LVLM models **102** with different checkpoints may introduce diversity, which may improve the generalization ability of LVLM **102**.
(67) By having the LLM **216** provide feedback for each response **308**, LVLM **102** may obtain a more comprehensive understanding of the strengths and weaknesses of VLM **112**.
(68) In an example, the described herein training paradigm of LVLM **102** may incorporate a fine-grained feedback modeling technique to specialize the evaluation of generated responses of LVLM **102**. The disclosed techniques may involve considering at least two aspects: helpfulness and honesty. As used herein, the term “helpfulness” refers to the relevance and usefulness of the generated response **308** to the given instruction **304** or question **312**. A helpful response **308** may provide accurate, informative, and relevant information.
(69) LLM evaluator (e.g., LLM **216**) of LVLM **102** may assess the helpfulness of generated responses **308** based on two primary criteria: 1) practical and beneficial information and 2) information that pertains exclusively to the user's question **312**. LLM **216** may evaluate whether the response **308** provides useful and relevant information about the image **310** that aligns with the query **312** of the user. In an example, such evaluation may involve assessing the accuracy and completeness of the content, and relevance to the needs of the user.
(70) As used herein, the term “honesty” refers to the extent to which a generated response (e.g., response **308**) is aligned with the visual content of the given image. The LLM **216** may evaluate honesty by determining whether the response **308** includes elements that do not exist in the image **310**.
(71) Generally, the LLM **216** may check if the response **308** describes visual elements that are not present in the image **310**. Hallucinated visual content may include, but is not limited to, objects, colors, or scenes that are not supported by the visual evidence.
(72) LVLM **102** may employ two different prompts to guide the LLM **216** in providing feedback on generated responses **308**. An example of the prompt for helpfulness evaluation may outline the scoring criteria and may instruct the LLM **216** to: generate reasons for the score, provide a rating, offer feedback for improvement. A similar prompt may be used for honesty evaluation, focusing on the alignment between the response **308** and the visual content of the image **310**.
(73) Some experiments show that generating reasons for scoring (chain-of-thought reasoning) may significantly improve the accuracy of the ratings of LLM **216**. In one implementation, the accuracy

may be improved because the LLM **216** may better articulate its reasoning process, leading to more consistent and reliable evaluations.

(74) In addition to scalar ratings, LLM **216** may also generate textual feedback to provide more detailed insights into the quality of the generated responses **308**. This textual feedback may be used in two ways: training phase and inference phase. Textual feedback may provide valuable information about the strengths and weaknesses of the response **308**. The LLM **216** may explicitly point out missing visual concepts or other shortcomings in the response **308**. During inference phase, textual feedback may be used to suggest ways to improve the generated response **308**. In other words, LVLM **102** may regenerate the response **308** by considering both the previous version and the feedback of LLM **216**. In one non-limiting example, textual feedback may provide more granular information than scalar ratings, allowing the LVLM **102** to learn more effectively. By incorporating textual feedback, LVLM **102** may generate responses **308** that are more accurate, relevant, and helpful.

(75) Additionally, the disclosed techniques may focus on three dimensions: harmlessness, helpfulness and honesty. Harmlessness may measure the potential for the response **308** to cause harm, such as, but not limited to, spreading misinformation or promoting harmful stereotypes. In an example, helpfulness may assess the usefulness and relevance of the response **308** to the given instruction **304** or question **312**. According to the disclosed techniques, honesty may evaluate the alignment between the response **308** and the visual content of the image **310**.

(76) In many traditional machine learning tasks, a reward model is used to guide the learning process. The reward model typically assigns a numerical value to each action or output, indicating how good or bad it is. The goal is to maximize the reward. Advantageously, the disclosed techniques essentially eliminate the need for an explicit reward model. Instead of a reward model, the LVLM **102** may be trained based on feedback generated by LLM **216**. As noted above, LLM **216** may provide critiques on the outputs of LVLM **102**, focusing on dimensions like harmlessness, helpfulness, and honesty. LVLM **102** may learn to adjust generated responses based on this feedback, gradually improving alignment of the generated responses with the desired qualities. Training a reward model may be expensive and time-consuming. By eliminating this step, the disclosed techniques become more cost-effective.

(77) By considering multiple dimensions, LVLM **102** may provide a more comprehensive evaluation of response quality.

(78) In this case evaluation of harmlessness by LVLM **102** may be decoupled from the image **310** content, allowing such evaluation to be outsourced to a specialized harmlessness detector. The disclosed techniques may have several advantages: efficiency, modularity and flexibility.

(79) By focusing on a specific task, the harmlessness detector may be optimized for efficiency and accuracy. The harmlessness detection component may be easily replaced or updated without affecting the rest of the model.

(80) The disclosed techniques may allow for the integration of different harmlessness detection techniques. In the illustrated example, LLM **216** may be used to perform the role of the harmlessness detector. The LLM **216** may be provided with a concrete task instruction to identify harmful content in the generated responses **308**. The disclosed techniques may leverage the ability of LLM **216** to understand and process language, making LLM **216** well-suited for this task. In an aspect, for verification, LLMs **216** may be adapted to detect a wide range of harmful content, including, but not limited to: hate speech, misinformation, and toxic language. In this example, LLMs **216** may be continuously improved by training LLMs **216** on new data and incorporating new techniques.

(81) During training, LVLM **102** may employ a two-stage training process. LLM **216** feedback data may be collected for each generated response **308**, including scalar ratings and textual feedback.

(82) The aforementioned attention to detail may help to identify and address potential issues in the generated responses **308**.

(83) As noted above, by incorporating human-centric evaluation criteria, feedback mechanism of LVLM **102** may help to ensure that the generated content aligns with human values and expectations.
(84) The comprehensive assessment process may help to identify and mitigate issues like hallucinations and unhelpful content, improving the overall quality of the generated responses **308**.
(85) Traditionally, in the context of LVLMs, resilience is important to ensure that the model is robust against malicious attacks and may maintain its performance in challenging environments. Adversarial prompting, a technique used to mislead or manipulate LLMs, is a significant threat to integrity and security of LVLMs. The disclosed techniques propose employing conditional Reinforcement Learning (RL). Conditional RL may allow LVLM **102** to learn to generate responses **308** that are conditioned on specific inputs, such as, but not limited to, instructions **304** or images **310**. As noted above, conditional RL may make LVLM **102** more adaptable and less susceptible to adversarial attacks. Conditional RL may be used to improve the performance of LVLM **102** on specific tasks, further enhancing resilience of LVLM **102**. Rejection sampling may involve rejecting generated responses **308** that are deemed to be low-quality or unsafe. Rejection sampling may help to maintain the quality and reliability of the output of LVLM **102**. Rejection sampling may be used as a defense mechanism against adversarial attacks. By rejecting malicious inputs, LVLM **102** may prevent the malicious inputs from influencing behavior of LVLM **102**. The combination of conditional RL and rejection sampling may provide LVLM **102** with a robust defense against adversarial prompting and other vulnerabilities. By rejecting low-quality responses, LVLM **102** may produce higher-quality output.
(86) RLAIF (Reinforcement Learning from AI Feedback) paradigm employed by LVLM **102** may offer a significant advancement in the field of LVLMs. By combining reinforcement learning, human-aligned fine-tuning, and extensive feedback, LVLM **102** may address several weaknesses found in conventional LVLM models.
(87) The techniques disclosed herein ensure that the responses **308** of LVLM **102** are aligned with human preferences and avoid harmful or misleading content. By incorporating human feedback and reinforcement learning, LVLM **102** may be better equipped to handle adversarial prompts and other threats to integrity of LVLM **102**. Training paradigm of LVLM **102** may be designed to be efficient and scalable, allowing for rapid development and deployment.
(88) FIG. **4** illustrates an example of an input image **402** and visual instruction **404** and an output response **406** that may be generated by LVLM model **102**, in accordance with the techniques of the disclosure.
(89) In this case, the generated response **406** may be helpful, honest and unharmful. In an aspect, to generate response **406**, LVLM **102** may need to be trained on more diverse and representative datasets, and responses of LVLM **102** may be carefully evaluated and refined to ensure the responses **406** align with human values and expectations. The training data **213** for many LVLMs may lack strong coherence between conversation turns, limiting the ability of LVLM to maintain context and engage in meaningful multi-turn interactions.
(90) FIG. **5** illustrates an example of an input image **502** and prompt **504** and a harmful output response **506** that may be generated by a LVLM model. More specifically, FIG. **5** illustrates responses **506** generated by LLaVA and Instruct BLIP models. Without a strong understanding of the conversation history, LVLMs may struggle to use prior context to improve their responses, leading to repetitive or irrelevant statements.
(91) Reinforcement Learning from Human Feedback (RLHF) is a common approach to align LLMs with human preferences. However, RLHF faces several challenges.
(92) RLHF involves collecting human ratings for a large number of generated responses, which may be time-consuming and expensive. Relying solely on human ratings may introduce biases and limitations due to the subjectivity of human judgment. To address the aforementioned challenges, LVLM **102** may employ novel techniques: Reinforcement Learning from AI Feedback (RLAIF).

The RLAIF may leverage an LLM to provide feedback on generated responses, eliminating the need for direct human ratings.

(93) RLAIF may be more efficient than traditional RLHF, as RLAIF does not require constant human input. RLAIF may be applied to large-scale models and datasets. By using LLM **216** for feedback, LVLM **102** may potentially mitigate some of the biases inherent in human judgment.

(94) One of the primary challenges in improving the interaction ability of LVLMs is the lack of coherence between utterances in multi-turn interactions within existing datasets. This limitation may hinder the ability of LVLMs to maintain context and engage in meaningful conversations. LVLM **102** may address the aforementioned challenge by categorizing natural language feedback into two distinct categories. Critique category may encompass feedback that highlights shortcomings or errors in the generated response. Refinement category may include feedback that suggests improvements or modifications to the response.

(95) By categorizing feedback in this way, LVLM **102** may better understand the specific areas where the model needs to improve and tailor its training accordingly. In an example, LVLM **102** may also employ a generate-annotate framework to systematically create useful multi-turn interaction data.

(96) The LLM **116** may generate responses to a given prompt or question. Human annotators may categorize the generated responses into critique and refinement categories. The aforementioned process may be repeated iteratively, with the LLM **116** learning from the annotations and generating improved responses over time.

(97) LVLM **102** may integrate Reinforcement Learning from AI Feedback (RLAIF) to address the limitations of existing LVLMs. By incorporating natural language feedback from LLMs **216**, LVLM **102** may improve at least two aspects. In an example, the LLM **216** may provide specific and actionable feedback on generated responses **308**, guiding the LVLM **102** towards more human-aligned outputs. LVLM **102** may use the feedback provided by LLM **216** to iteratively refine responses **308**, ensuring the responses **308** are consistent with human values and expectations. Natural language feedback may help LVLM **102** better understand the context of conversations, leading to more coherent and engaging interactions. By analyzing feedback, LVLM **102** may adapt the responses **308** to different users and situations, fostering more personalized and satisfying interactions. RLAIF may eliminate the need for human annotators, making the training process more efficient and scalable. LLM **216** may provide feedback in real-time, allowing LVLM **102** to continuously improve performance.

(98) For example, LVLM **102** may perform a categorization of NLF into two primary categories. Critique feedback may highlight the specific strengths and weaknesses of the response **308** generated by LVM **112**. Critique feedback may include identifying areas where the response is informative, accurate, or relevant, as well as areas where the response **308** (e.g., responses **506** shown in FIG. **5**) is lacking or incorrect. Refinement feedback may offer specific suggestions on how the LVLM **102** may improve its response. In other words, refinement feedback may involve providing concrete examples, suggesting alternative phrasings, or pointing out missing information. Additionally, by carefully categorizing NLF, LVLM **102** may transform the NLF into more coherent multi-turn interactions, further enhancing the interaction capabilities of LVM **112**.

(99) LVLM **102** may analyze the critique and refinement feedback to identify common patterns or themes. LVLM **102** may develop new prompts or questions based on the identified patterns, focusing on areas where the LVLM **102** struggled or could improve.

(100) Natural Language Feedback (NLF) may provide valuable insights into the quality of generated responses. By incorporating NLF into the conditional RL algorithm, LVLM **102** may significantly enhance the ability of the LVLM **102** to generate human-aligned and informative outputs. To effectively incorporate NLF into the conditional RL algorithm, LVLM **102** may need to be trained to generate responses **308** that are conditioned on the specific NLF provided. The NLF may be converted into a suitable representation that may be processed by the LVLM **102** (e.g.,

embedding). The NLF representation may be concatenated with the input prompt **312** or image **310** to create a combined input. In this context, LVLM **102** may be trained to generate responses **308** based on this combined input. LVLM **102** may learn to distinguish between good and bad responses **308** through the reward function used in the RL algorithm. By associating positive rewards with high-quality responses **308** and negative rewards with low-quality responses **308**, LVLM **102** may gradually learn to generate more desirable outputs.
(101) Meta-learning may be employed to further enhance the ability of LVLM **102** to generate high-quality responses **308**. Meta learning may involve training LVLM **102** to learn how to learn from new data and adapt to different scenarios. By meta-learning, LVLM **102** may become more flexible and adaptable, improving its performance over time. By incorporating NLF, LVLM **102** may better align responses of LVLM **102** with human expectations. NLF may provide valuable context that helps the model generate more relevant and informative responses.
(102) FIG. **6** is a detailed block diagram illustrating an example of Natural Language Feedback annotation performed by a LLM, in accordance with the techniques of the disclosure. As shown in FIG. **6**, LVLM **102** may receive a combined input. The combined input may include an image **602** (e.g., a visual representation of a scene or object) and a prompt/query **604** (a textual description or question related to the image **602**). In an aspect, the LVLM **102** may process the combined input, combining the visual and textual information to create a unified representation, as described above. Based on this representation, the LVLM **102** may generate a response **606**, such as response **606**A (e.g., "I see a happy dog flying with kids."). The LLM **216** may be trained on dense annotations of the images **608** as input, to provide additional context and information. As shown in FIG. **6**, the LLM **216** may generate Natural Language Feedback (NLF) consisting of: numerical score **610**, critique **612** and refinement **614**. Numerical score **610** may be a quantitative rating of the response **608** of LVLM **102** (e.g., 2 out of 5). Critique **612** may be specific feedback on the shortcomings of the response **606** (e.g., "hallucinating non-existing dogs"). Refinement **614** may include, but is not limited to, suggestions for improving the response **606** (e.g., "Change the dog to Doraemon and focus more on descriptions of surroundings"). In this example, the LVLM **102** may use this feedback **610-614** to improve its future responses.
(103) In an example, during pretraining, 8 million image-caption pairs may be used as input. The goal of pretraining may be to train a strong captioner LVLM **102** that may accurately describe images. During SFT, for example, LLaVA dataset (160,000 samples) may be used. The objective of SFT may be to align the linear projector **114** of LVLM **102** between visual and textual features, making the LVLM **102** capable of following instructions.
(104) A well-trained LVLM **102** may generate accurate captions for images.
(105) To collect feedback on performance of LVLM **102**, a subset of, for example, 30,000 samples may be selected from the LLaVA dataset. This subset may be carefully filtered to ensure no duplicate images and that questions are answered solely based on the image. The subset containing no duplicate images may prevent LVLM **102** from learning biases or shortcuts based on repeated images.
(106) FIG. **7** is a flowchart illustrating an example mode of operation for a machine learning system, according to techniques described in this disclosure. Although described with respect to computing system **200** of FIG. **2** having processing circuitry **243** that executes machine learning system **204**, mode of operation **700** may be performed by a computing system with respect to other examples of machine learning systems described herein.
(107) In mode of operation **700**, processing circuitry **243** executes machine learning system **204**. Machine learning model **204** may providing a plurality of visual queries to LVLM **102** (**702**). In an example, the visual query may include at least an image and a textual query related to the image. Next, LVLM **102** may process the plurality of visual queries to extract one or more visual embeddings from each of the plurality of visual queries (**704**). The LVLM **102** may include a Visual Language Model (VLM) **112**, a first Large Language Model (LLM) **116** and a linear

projection layer **114** interconnecting the VLM **112** and the LLM **116**. The VLM **112** may process the input image and may extract one or more visual features of the input image. LVLM **102** may generate a response to the visual query based on the generated one or more visual embeddings (**706**). In an aspect, machine learning system **204** may use human-annotated dense captions as ground truth examples to evaluate the generated responses of LVLM **102**. A second LLM **216** may evaluate the generated response to verify that the generated response satisfies one or more predefined criteria (**708**). In an example, such evaluation may involve assessing the accuracy and completeness of the content, and relevance to the needs of the user. Next, the second LLM may provide a feedback to the LVLM, in response to the evaluating the generated response (**710**). This textual feedback may be used in two ways: a training phase and an inference phase. Textual feedback may provide valuable information about the strengths and weaknesses of the response **308**. The LLM **216** may explicitly point out missing visual concepts or other shortcomings in the response **308**. During inference phase, textual feedback may be used to suggest ways to improve the generated response **308**. It should be noted that **706-710** may be repeated by the machine learning model **204** for each of the plurality of visual queries. The machine learning model **204** may fine-tune the LVLM **102** using aggregated feedback provided by LLM **216** for the plurality of visual queries (**712**). The generated feedback may offer specific suggestions on how the LVLM **102** may improve its response. In other words, the generated feedback may involve providing concrete examples, suggesting alternative phrasings, or pointing out missing information.

(108) The techniques described in this disclosure may be implemented, at least in part, in hardware, software, firmware or any combination thereof. For example, various aspects of the described techniques may be implemented within one or more processors, including one or more microprocessors, digital signal processors (DSPs), application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), or any other equivalent integrated or discrete logic circuitry, as well as any combinations of such components. The term “processor” or “processing circuitry” may generally refer to any of the foregoing logic circuitry, alone or in combination with other logic circuitry, or any other equivalent circuitry. A control unit comprising hardware may also perform one or more of the techniques of this disclosure.

(109) Such hardware, software, and firmware may be implemented within the same device or within separate devices to support the various operations and functions described in this disclosure. In addition, any of the described units, modules or components may be implemented together or separately as discrete but interoperable logic devices. Depiction of different features as modules or units is intended to highlight different functional aspects and does not necessarily imply that such modules or units must be realized by separate hardware or software components. Rather, functionality associated with one or more modules or units may be performed by separate hardware or software components or integrated within common or separate hardware or software components.

(110) The techniques described in this disclosure may also be embodied or encoded in computer-readable media, such as a computer-readable storage medium, containing instructions. Instructions embedded or encoded in one or more computer-readable storage mediums may cause a programmable processor, or other processor, to perform the method, e.g., when the instructions are executed. Computer readable storage media may include random access memory (RAM), read only memory (ROM), programmable read only memory (PROM), erasable programmable read only memory (EPROM), electronically erasable programmable read only memory (EEPROM), flash memory, a hard disk, a CD-ROM, a floppy disk, a cassette, magnetic media, optical media, or other computer readable media.

## Claims

1. A method for fine-tuning a Large Visual Language Model (LVLM), the method comprising: providing a plurality of visual queries, wherein each of the plurality of visual queries comprises at least an image and a textual query related to the image; processing, by the LVLM, the plurality of visual queries to extract one or more visual embeddings from each of the plurality of visual queries, wherein the LVLM comprises a Visual Language Model (VLM), a first Large Language Model (LLM), and a linear projection layer interconnecting the VLM and the LLM; for each of the plurality of visual queries: i) generating, by the LVLM, a response to the corresponding visual query based on the corresponding one or more visual embeddings; ii) evaluating, by a second LLM, the generated response to verify that the generated response satisfies one or more predefined criteria; and iii) providing, by the second LLM, a feedback to the LVLM, in response to the evaluating the generated response; and fine-tuning the LVLM using aggregated feedback provided by the second LLM for the plurality of visual queries.
2. The method of claim 1, wherein the one or more predefined criteria comprise at least one of helpfulness, honesty, and harmlessness.
3. The method of claim 1, wherein the feedback comprises a Natural Language Feedback (NLF).
4. The method of claim 3, wherein the feedback comprises at least a numerical score, critique feedback, and refinement feedback.
5. The method of claim 4, wherein the refinement feedback suggests improvements or modifications to the generated response.
6. The method of claim 3, further comprising training the LVLM using the NLF.
7. The method of claim 6, wherein training the LVLM further comprises: training the LVLM using the NLF incorporated into a conditional Reinforcement Learning (RL) algorithm.
8. A computing system for fine-tuning a Large Visual Language Model (LVLM), the computing system comprising: processing circuitry in communication with storage media, the processing circuitry configured to execute a machine learning system comprising the LVLM, the processing circuitry configured to: provide a plurality of visual queries, wherein each of the plurality of visual queries comprises at least an image and a textual query related to the image; process, by the LVLM, the plurality of visual queries to extract one or more visual embeddings from each of the plurality of visual queries, wherein the LVLM comprises a Visual Language Model (VLM), a first Large Language Model (LLM), and a linear projection layer interconnecting the VLM and the LLM; for each of the plurality of visual queries: i) generate, by the LVLM, a response to the corresponding visual query based on the corresponding one or more visual embeddings; ii) evaluate, by a second LLM, the generated response to verify that the generated response satisfies one or more predefined criteria; and iii) provide, by the second LLM, a feedback to the LVLM, in response to the evaluating the generated response; and fine-tune the LVLM using aggregated feedback provided by the second LLM for the plurality of visual queries.
9. The system of claim 8, wherein the one or more predefined criteria comprise at least one of helpfulness, honesty, and harmlessness.
10. The system of claim 8, wherein the feedback comprises a Natural Language Feedback (NLF).
11. The system of claim 10, wherein the feedback comprises at least a numerical score, critique feedback, and refinement feedback.
12. The system of claim 11, wherein the refinement feedback suggests improvements or modifications to the generated response.
13. The system of claim 10, the processing circuitry further configured to: train the LVLM using the NLF.
14. The system of claim 13, wherein the processing circuitry configured to train the LVLM is further configured to: train the LVLM using the NLF incorporated into a conditional Reinforcement Learning (RL) algorithm.
15. Non-transitory computer-readable storage media having instructions encoded thereon, the

instructions configured to cause processing circuitry to: provide a plurality of visual queries, wherein each of the plurality of visual queries comprises at least an image and a textual query related to the image; process, by a Large Visual Language Model (LVLM), the plurality of visual queries to extract one or more visual embeddings from each of the plurality of visual queries, wherein the LVLM comprises a Visual Language Model (VLM), a first Large Language Model (LLM), and a linear projection layer interconnecting the VLM and the LLM; for each of the plurality of visual queries: i) generate, by the LVLM, a response to the corresponding visual query based on the corresponding one or more visual embeddings; ii) evaluate, by a second LLM, the generated response to verify that the generated response satisfies one or more predefined criteria; and iii) provide, by the second LLM, a feedback to the LVLM, in response to the evaluating the generated response; and fine-tune the LVLM using aggregated feedback provided by the second LLM for the plurality of visual queries.
16. The storage media of claim 15, wherein the one or more predefined criteria comprise at least one of helpfulness, honesty, and harmlessness.
17. The storage media of claim 15, wherein the feedback comprises a Natural Language Feedback (NLF).
18. The storage media of claim 17, wherein the feedback comprises at least a numerical score, critique feedback, and refinement feedback.
19. The storage media of claim 18, wherein the refinement feedback suggests improvements or modifications to the generated response.
20. The storage media of claim 17, the instructions further configured to cause processing circuitry to: train the LVLM using the NLF.